BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 6197

Retyped faithfully from the original board paper — same questions, no changes.

Paper: I Objective-(iv)

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Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6197

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جوابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. $(A \cup B)^c$ = ...

(A) $A \cup B$ (B) $A \cap B$

(C) $A^c \cup B^c$ (D) $A^c \cap B^c$

Q2. The product of the roots of equation $x^2+2x+1=0$, is .....

(A) 2 (B) 3

(C) 1 (D) -1

Q3. If $4^x=1/2$ then x= .....

(A) 1/4 (B) 1/2

(C) $-1/2$ (D) 2

Q4. Rank of the matrix [1 0; 0 1] is.....

(A) 2 (B) -1

(C) 0 (D) $\sqrt{(-1)}$

Q5. In any triangle ABC, $(c^2+a^2-b^2)/2ac$ = ...

(A) cosα (B) cosβ

(C) cosγ (D) cos(β+α)

Q6. A⊆B then complement of A in B (B-Universal):

(A) A−B (B) B−A

(C) A∩B (D) A∪B

Q7. A/(x−1) + B/(x+1) is a partial fraction of:

(A) $1/(x^3-1)$ (B) $1/(x^2-1)$

(C) $1/(1-x^2)$ (D) $1/(x^2+1)$

Q8. 2sin45°+1/2cosec45° = ...

(A) 1 (B) 1

(C) 3/√2 (D) √(2/3)

Q9. Multiplicative inverse of (0,-1) ∈ C, is:

(A) (0,1) (B) (1,0)

(C) (−1,0) (D) (1,1)

Q10. ${}^6P_3$ = ...

(A) 6 (B) 18

(C) 36 (D) 120

Q11. If cos x=1/√2, then reference angle is:

(A) π/6 (B) π/4

(C) π/3 (D) π/2

Q12. The value of $\sec(\sin^{-1}(\sqrt{3}/2))$ = ...

(A) √3/2 (B) 1/√2

(C) 2 (D) 1/2

Q13. If |k 4; 4 k| = 20 then k = ...

(A) ±36 (B) ±24

(C) ±16 (D) ±6

Q14. Range of sin(x/2) is:

(A) [1/2,−1/2] (B) [−2,2]

(C) [2,−2] (D) [−1,1]

Q15. $\sin(-300°)$ = ...

(A) $-\sqrt{3}/2$ (B) $\sqrt{3}/2$

(C) $2/\sqrt{3}$ (D) 0

Q16. (Z,+) has identity element:

(A) 0 (B) i

(C) [illegible in source scan] (D) -1

Q17. (r+1)th term in the expansion of $(a+b)^n$ is:

(A) (n r) a^{n-r} b^r (B) (n r) a^{n-r} b^{r-1}[illegible in source scan]

(C) (n r) a^{n+r} b^r (D) (n r) a^{n+r} b^{r+1}

Q18. Every non-recurring, non-terminating decimal represents ..... number.

(A) rational (B) irrational

(C) whole (D) natural

Q19. If a=1, b=5 then A×H = ...

(A) 2/5 (B) 5/2

(C) 5 (D) -5

Q20. If $a_{n-2} = 3n-11$, then nth term is:

(A) 3n+2 (B) 3n−5

(C) 3n+5 (D) 3n−3

Paper: I Subjective

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Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section B is compulsory. Attempt any THREE questions from section C.

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Factorize $9a^2+16b^2$.

(ii) Define modulus of a complex number.

(iii) Find the multiplicative inverse of $(-4,7)$.

(iv) Express the complex number $1+i\sqrt{3}$ in polar form.

(v) Write the set $\{x|x\in Q \wedge x^2=2\}$.

(vi) Convert $(A\cap B)'=A'\cup B'$ into logical form.

(vii) Define diagonal matrix and give an example.

(viii) Show that for a non-singular matrix A, $(A^{-1})^{-1}=A$.

(ix) Define co-factor of an element.

(x) Solve x^(1/2)−x^(1/4)−6=0

(xi) Reduce $x^{-2}-10=3x^{-1}$ to quadratic form.

(xii) Show that $x^3-y^3=(x-y)(x-\omega y)(x-\omega^2 y)$.

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Define improper fraction.

(ii) Change $(x^2+1)/(x^2-1)$ into proper fraction.

(iii) Find 9th term of the sequence −1/5, −1/3, −1, ...

(iv) For a=2i, b=4i, show that $G^2=A\times H$.

(v) Find the first term of the geometric series if $a_n=(-3)(2/5)^n$.

(vi) If 1/a, 1/b, 1/c are in G.P, show that the common ratio is $\pm\sqrt{(a/c)}$.

(vii) If ${}^nC_8={}^nC_{12}$, find n.

(viii) How many triangles and diagonals can be formed by joining the vertices of 8-sided polygon?

(ix) Define circular permutation.

(x) From the expansion of $(3x/2 - 1/3x)^{11}$, find the sixth term from the end.

(xi) Expand (8−5x)^(2/3) up to four terms.

(xii) Evaluate $\sqrt[3]{31}$ correct to three decimal places.

Q4. (Marks 18) Write short answers to any NINE parts. (9x2=18)

(i) Express $\theta=120°40'$ in radians.

(ii) If $\sin\theta=12/13$ and terminal arm of angle is in quadrant I, find $\tan\theta$ and $\cos\theta$.

(iii) Find the value of $\sin(-300°)$.

(iv) Prove that $\sin(\theta+\pi/6)+\cos(\theta+\pi/3)=\cos\theta$.

(v) Write down the half angle identity for $\tan(\alpha/2)$.

(vi) Define period of a trigonometric function.

(vii) Prove that period of sin function is 2π.

(viii) Write down the domain and range for y=tan x.

(ix) Solve the right triangle ABC in which $\gamma=90°$, $a=3.28$, $b=5.74$.

(x) Write half angle formulas $\sin(\gamma/2)$ and $\cos(\gamma/2)$.

(xi) Define and draw an oblique triangle.

(xii) Find the value of $\sin(\cos^{-1}(\sqrt{3}/2))$.

(xiii) Find the solution $x\in[0,2\pi]$, $\sin x=-\sqrt{3}/2$.

> Note: Attempt any THREE questions. Each question carries (5+5=10) marks.

Q5. (Marks 10)

(a) [5] Solve the system of linear equations by Cramer's Rule. $2x_1-x_2+x_3=8$; $x_1+2x_2+x_3=6$; $x_1-2x_2-x_3=$[illegible in source scan]

(b) [5] Solve systems of equations: $x^2+y^2=5$; $x^2+2y^2=17$

Q6. (Marks 10)

(a) [5] Resolve $(x^2-10x+13)/((x-1)(x^2-5x+6))$ into partial fractions.

(b) [5] Show that the sum of 'n' A.Ms between a and b is equal to n times their A.M.

Q7. (Marks 10)

(a) [5] Prove that ${}^nC_r+{}^nC_{r-1}={}^{n+1}C_r$.

(b) [5] If x is nearly equal to 1 then prove that $px^p-qx\hat{}q=(p-q)x\hat{}(p+q)$.

Q8. (Marks 10)

(a) [5] Prove that $(\tan\theta+\sec\theta-1)/(\tan\theta-\sec\theta+1) = \tan\theta+\sec\theta$.

(b) [5] Prove that $\tan^{-1}(120/119) = 2\cos^{-1}(12/13)$.

Q9. (Marks 10)

(a) [5] If $\alpha+\beta+\gamma=180°$, show that $\cot\alpha\cot\beta+\cot\beta\cot\gamma+\cot\gamma\cot\alpha=1$.

(b) [5] Prove that $r_1+r_2+r_3-r=4R$.